DOCKHUB · DESIGN DECISION RECORD

DockHub TRL 2 review decisions

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	Items 10 and 11 decided by Amish as recommended (DKH-DEC-001)
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	Record the TRL 2 review items adopted for TRL 3 under Amish's 2026-09-25 instruction, and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** items 1 to 9, 12 and 13 decided by Amish, 2026-09-25: go with recommendation (see DKH-DDR-002). Items 10 and 11 had no recommendation at TRL 2; recommendations were written for them later, and Amish approved them on 2026-10-02: "i approve your recommendations for all 555 open decisions." (DKH-DEC-001).

Context

The TRL 2 review note (`docs/REVIEW.md`, session of 2026-09-25) listed twelve points as "Proposed, awaiting Amish". On 2026-09-25 Amish asked for this batch of repos to be taken through the usual process with the instruction "you know the drill, nothing gets past TRL 3". He has not reviewed the DockHub points item by item. Under that instruction, every point that carried a recommendation is adopted as recommended so that the TRL 3 calculations, model and drawing can proceed, and each stays open for his review. Points with no recommendation stay "Proposed, awaiting Amish". TRL 4 is on hold by Amish's instruction.

Options considered

The options for each item are in `docs/REVIEW.md` (TRL 2 session) and DKH-PRC-001 v0.2, "Key design choices". They are not repeated here.

Decision

Table 1. Items adopted for TRL 3.

#	Item	Status	Where it now lives
1	Budget (R16): build the four-bay cabinet and fit two bays for the first prototype (option a)	Decided by Amish, 2026-09-25: go with recommendation (DKH-DDR-002). `budget_usd` stays at $1,200 and now covers the four-bay cabinet with bays 1 and 2 fitted; SwapCell packs stay excluded	DKH-REQ-001 v0.3 R16, `bom/bom.csv`, `cad/src/model.py` (`bays_fitted`), DKH-DWG-001
2	Solar target (R6): keep the 400 W canopy as shade and supplement, and relax R6 from 20 % to 10 %	Decided by Amish, 2026-09-25: go with recommendation (DKH-DDR-002)	DKH-REQ-001 v0.3 R6, DKH-PRC-001 v0.3
3	Four bays in one row at hand height rather than eight in two rows	Decided by Amish, 2026-09-25: go with recommendation (DKH-DDR-002)	DKH-PRC-001 v0.3, `cad/src/model.py`
4	One certified charger and one CAN channel per bay rather than a shared charger matrix	Decided by Amish, 2026-09-25: go with recommendation (DKH-DDR-002)	DKH-PRC-001 v0.3, `bom/bom.csv` lines 5 and 6
5	Separate steel compartment per bay with a rear plenum venting through the roof	Decided by Amish, 2026-09-25: go with recommendation (DKH-DDR-002)	DKH-PRC-001 v0.3, `cad/src/model.py`
6	Bay doors stay locked during a fire alarm	Decided by Amish, 2026-09-25: go with recommendation (DKH-DDR-002)	DKH-PRC-001 v0.3
7	Solar through an MPPT controller to bay 1 rather than a hybrid inverter	Decided by Amish, 2026-09-25: go with recommendation (DKH-DDR-002)	DKH-PRC-001 v0.3, `bom/bom.csv` line 13
8	Offline-first NFC access with an optional app	Decided by Amish, 2026-09-25: go with recommendation (DKH-DDR-002)	DKH-PRC-001 v0.3
9	State-of-health gate at 70 % for releasing packs to riders	Decided by Amish, 2026-09-25: go with recommendation (DKH-DDR-002)	DKH-PRC-001 v0.3

The TRL 2 review checked the `project.yaml` pitch and problem lines against the sources and left them unchanged; it did not recommend a new wording, so nothing was applied to `project.yaml` or `README.md` under this record.

Items that remain open, and engineering proposals

Table 2. Open items and TRL 3 engineering proposals.

#	Item	Status
10	Pack ownership model: station-owned pool, rider-owned one-for-one exchange, or fleet packs	Decided by Amish, 2026-10-02, as recommended: a station-owned pool of packs exchanged one for one, as the concept and R10 assume; to be confirmed in co-design (DKH-DEC-001)
11	First pilot site and partner: a delivery-worker group in a high-income city, or a moto-taxi hub in East Africa	Decided by Amish, 2026-10-02, as recommended: a delivery-worker group in a high-income city, New York City first, with Los Deliveristas Unidos (Workers Justice Project) as the first candidate partner to approach (DKH-DEC-001)
12	Building to SwapCell interface v0.3 instead of v0.2 (coded INTERLOCK receptacle, item W; dock host type 1; class D catch, item V). SwapCell issued v0.3 on 2026-09-25 with Amish's approval of its additions; DockHub adopts it because a v0.3 pack stays dead in a receptacle without the 10 kΩ coding resistor	Decided by Amish, 2026-09-25: go with recommendation (DKH-DDR-002)
13	TRL 3 engineering proposals in DKH-CAL-001: release a pack at 80 % state of charge or more; chargers with a power factor of 0.9 or better; a flush plinth (1,000 x 500 mm) to keep the footprint inside R13; a 15 min rider patience and the demand profile used in the throughput model	Decided by Amish, 2026-09-25: go with recommendation (DKH-DDR-002)

Consequences

- R6 is now judged against 10 %. DKH-CAL-001 finds 9.3 % with the service-first release rule, so **R6 is still not met**; routing flat packs into bay 1 in daylight reaches 11.4 % but turns one rider in 20 away at the dinner peak.
- R16 is judged on the two-bay fit: $1,199 against $1,200 (at risk). A two-bay prototype holds only one resident pack, because one bay must always be empty to receive a returned pack. It serves 9 of the 20 duty-case swaps, so it demonstrates the station but cannot run the duty case; R4 is met only by the four-bay fit ($1,423).
- The budget figure in `project.yaml` is unchanged. Option (b) of item 1 ($1,450) was not recommended and is not adopted.
- DockHub cites SwapCell interface v0.3 (item 12). Nothing in the SwapCell interface is changed by DockHub.
- On 2026-09-25 Amish accepted all recommendations ("i accept all your recommendations, go with them across all repos"). Items 1 to 9, 12 and 13 are decided; items 10 and 11 had no recommendation and stay open. See DKH-DDR-002.